Machine Learning

Final Exam

Aleksandr Petiushko

ML Research

Content

Content

Exam information

- Exam format: **hand-written on paper**
- Exam duration: **30 min**
- Materials allowed to be used: **hand-written notes (1 sheet of paper, both sides can be used)**
- **No one** is allowed to use either the **phone**, **laptop** or **printed materials**
- Exam parts:
  1. Short questions about the course content: 50%
  2. Simple ML-related calculations: 50%

Exam process

Exam procedure:

- Exam Assignment will be open in **Canvas** for the **time of Exam** (and **only** during this time!)
- You'll need to write Exam Assignment on paper sheets and when you finish, take a photo / convert to pdf and **submit** as a 'File Upload' to Canvas
- You'll have only **1 attempt** to submit, so please be careful!
- After submission, please **deliver** the written sheets to **Professor** / **Classroom Manager** (to be collected, stored, and probably scanned for better quality)

Proctor Policy I

- All students taking in-person exams are expected to follow academic integrity standards to ensure a fair and secure testing environment. Only materials explicitly permitted by the professor may be used during the exam. Any authorized materials, including cheat sheets, must be clearly defined by the professor in advance (for example, one handwritten page) and may be briefly checked at the start of the exam to ensure compliance.
- Students arriving late may be admitted during a limited grace period, typically 10–15 minutes after the exam begins. After that time, entry may be restricted in order to maintain exam integrity and minimize disruption. Students who arrive late will not receive additional testing time.
- All personal belongings, including bags, phones, smartwatches, headphones, and unauthorized notes, must be placed in a designated area before the exam begins. The designated area may include an adjacent seat or underneath the student's seat, as directed by the professor/proctor. All electronic devices must be completely powered off, not simply silenced or placed on vibrate mode.

Proctor Policy II

- Students must follow all instructions provided by the instructor or proctor and remain seated unless permission is granted to leave the room. Communication with other students is prohibited at all times during the exam.
- Short restroom breaks may be permitted one student at a time at the discretion of the professor/proctor. Students may not take any exam materials, notes, or electronic devices with them during a break. Exam time will not be paused or extended for restroom breaks.
- Any behavior suggesting academic misconduct, including but not limited to looking at another student's work, communicating during the exam, accessing unauthorized materials, or attempting to use concealed electronic devices, will be treated as suspected cheating. Suspected violations will be documented and reported in accordance with institutional academic integrity policies and procedures. Consequences for confirmed violations may include disciplinary action as outlined by the institution.

Remarks on process

- You need to add your name at the top left of the each side of every sheet
- You need to add the page number at the top right of the each side of every sheet (1, 2, 3, 4, 5, ...)
- Allowed: Only one person at a time may leave the room

Exam topics: ML concepts

Demonstrate the deep knowledge of the following ML concepts:

- Supervised Learning, types of models (high-level)
- Input feature types and dimensionality
- Empirical vs Structural Risk Minimization
- Overfitting vs Underfitting and methods to avoid them
- Cross-Validation
- Model Selection pipeline and why it is important
- Classification vs Regression
- Classification and Regression loss functions
- Classification quality metrics (including accuracy, precision, and recall)
- Regression quality metrics (including MAE, MSE, and RMSE)
- Binary vs Multi-class Classification
- Micro- vs Macro- Averaging for Multi-class Classification
- L1 (Manhattan) and L2 (Euclidian) norms (distances)
- k-NN Classification and k-NN Regression
- Linear Regression: Ridge, LASSO, and Elastic variants

Exam topics: ML calculations

Additionally, to be able to compute auxiliary things like:

- TP, FP, FN, TN
- TPR, FPR, FNR, TNR
- MAE, MSE, RMSE
- Accuracy, Precision, Recall
- Empirical Risk
- L1 (Manhattan) and L2 (Euclidian) norms (distances), and simple equalities/inequalities based on them

Thank you *all*!